Foundations of Cybersecurity/Applied Cryptography
13 June 2023

Exercise n. 1 [12] [All]

1. Illustrate the Diffie-Hellman protocol.
2. Argue about its security w.r.t. a passive and an active adversary.
3. Argue why it is more secure to run the protocol in a sub-group H of $\mathbb{Z}_p^*$ rather than in the $\mathbb{Z}_p^*$ itself. Which feature must H possess?

Exercise n. 2 [10] [All]

Consider the following authentication and key agreement protocol:

M1 $A \rightarrow B: \{A, B, n_A, K'\}_{K_{AB}}$

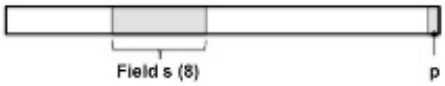
M2 $B \rightarrow A: \{A, B, n_A\}_{K'}$,

where K_{AB} is a long-term key shared by Alice and Bob, n_A is a nonce, and K' is the session key.

1. Analyse the protocol and specify the assumptions under which it fulfils the key authentication and key confirmation requirements.
2. Discuss whether n_A is necessary or if it can be removed from the protocol.
3. Argue whether the protocol is subject to a replay attack.
4. In case of positive answer to the previous question, modify the protocol to remove the vulnerability.

Exercise n. 3 [8] [2022-23]

Assume that a plaintext P has the format specified in the figure where, in particular, *s* is an 8-bit field that specifies an amount of money and *p* is a parity bit s.t. *p* is 0 if the number of 1s in the plaintext (bit p excluded) is even; it is 1 otherwise.



The whole plaintext is encrypted by means of one-time-pad. Answer the following questions.

1. Is this encryption scheme malleable?
2. Assume that field *s* specifies the value 130. Argue whether and how, it is possible to modify the cipher-text so that the decrypted plaintext specifies 146 in the field s and such a modification goes undetected.
3. Propose a possible countermeasure.
4.

Exercise n. 3 [8] [earlier]

Find and explain the vulnerabilities of the following function and patch them.

```
unsigned int do_stuff(unsigned int a, int b){
  return a<<b;
}
```

Solution

EXERCISE N.1
See theory.

EXERCISE N.2
Question 1. The main assumption is that B believes K' fresh.
Question 2. If A generates fresh session keys, then n_A may be removed.
Question 3. Because of the assumption at point 1), an adversary that manages to obtain a session key K' then (s)he can impersonate Alice with respect to Bob.
Question 4. A possible solution is the following:
M1 $B \rightarrow A: n_B$
M2 $A \rightarrow B: \{A, B, n_B, K'\}_{K_{AB}}$
M3 $B \rightarrow A: \{B, A, n_B + 1\}_{K'}$

EXERCISE N.3 [2022-23]
Question 1. The encryption scheme is malleable. A simple way to prove it is the following. Let P[i], C[i], and K[i] be the i-th bit of the plaintext, ciphertext and key, respectively, s.t. C[i] = P[i] xor K[i]. Notice that P[0] = p. Finally let C' be the modified ciphertext and P' the resulting plaintext after decryption. Notice that an adversary can easily complement a bit of the plaintext by operating on the ciphertext. Assume that the adversary wishes to complement bit i. Then, (s)he computes C'[i] = C[i] xor 1 = (P[i] xor K[i]) xor 1 = (P[i] xor 1) xor K[i] = P'[i] xor K[i]. If follows that P'[i] = P[i] xor 1, that is, P'[i] is the complement of P[i] (P'[i] = ~P[i]).

However, for the attack to go undetected, the parity bit must be consistently modified as well. Notice that since P[i] is complemented the number of 1 either increments or decrements by one. In both cases the parity bit must be complemented as well, that is C'[0] = C[0] xor 1.
Question 2. The attack consists in complementing both C[0] and s[4]:

$$s' \leftarrow s \oplus 00010000;\ p' \leftarrow p \oplus 1$$

Question 3. The problem can be solved by replacing the parity bit by a tag resulting from a secure hash function.

EXERCISE N.3 [EARLIER]
This is a case of usual conversions where operation mixes signed and unsigned operands. In this case, the better-ranked-unsigned rule applies and int in converted to uint. The problem appears when the int operand is negative. A possible sanitization is the following:

```
unsigned int do_stuff(unsigned int a, int b){
    if (b < 0) return a << -b;
  return a<<b;
}
```